

Ternary Logic and Logic with Independence

This note explores a possible relationship between a computational ternary logic and a richer logic that explicitly represents independence.

Computational ternary logic

Use three values: **T**: the statement has a verified proof. **F**: the negation has a verified proof. **U**: neither has been established. The value U represents only the current state of knowledge; it does not imply mathematical independence.

Logic with independence

A richer logic distinguishes: **T**: provable. **F**: refutable. **I**: independent of the chosen formal theory (under the required metatheoretic assumptions). **U**: computationally unresolved.

A projection between the logics

Define a projection π from the richer logic to the ternary logic by: $\pi(T)=T$ $\pi(F)=F$ $\pi(I)=U$ $\pi(U)=U$
Under this projection, independence information is intentionally forgotten. Every proved theorem and every proved refutation remains valid after projection, while both genuine independence and unresolved search become the single value U.

Research question

Rather than immediately calling this a transfer principle, a precise mathematical goal is to investigate whether there exists an embedding or structure-preserving map between the richer logic with independence and the computational ternary logic. Such a theorem would identify exactly which reasoning steps are preserved and which information is lost.

Application to Predator

Predator could use ternary logic internally for efficient proof-search control while the higher-level Automated Logical Decider records additional metatheoretic information, including formally established independence results. This separates computational efficiency from logical expressiveness while preserving soundness through formal verification.